

CPSC 544

Advanced Software Process

Part I: Software Process Maturity

1. A Software Maturity Framework

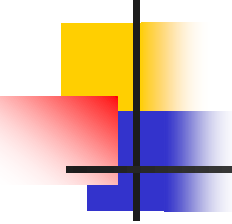
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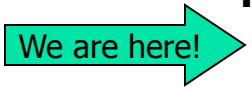
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Outlines



We are here!

Part I: Software Process Maturity

1. A Software Maturity Framework
2. The Principles of Software Process Change
3. Software Process Assessment
4. The Initial Process

Part II: The Repeatable Process

5. Managing Software Organizations
6. The Project Plan
7. Software Configuration Management (Part I)
8. Software Quality Assurance

Part III: Defined Process

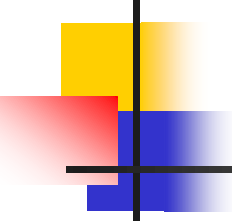
9. Software Standards
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14. The Software Engineering Process Group

Part IV: The Managed Process

15. Data Gathering and Analysis
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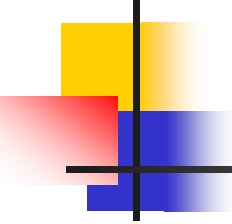
Part V: The Optimizing Process

17. Defect Prevention
18. Automating the Software Process
19. Contracting for Software
20. Conclusion



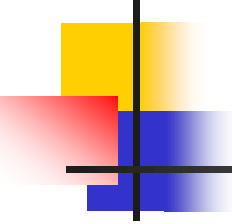
Introduction

- From this class, you will learn how to deal with the following questions:
 - What is a **software development process**?
 - How can I develop software through a certain **process**?
 - How good is my current software **process**?
 - What must I do to **improve** it?
 - Where do I **start**?



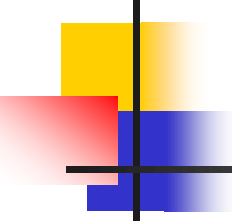
Introduction

- This lecture is based on the textbook and references.
 - The textbook describes about a generic software process.
 - Basis for CMM
 - References
 - ISO 12207 Software Lifecycle Processes
 - ISO 15504 Process Assessment (SPICE)
 - CMMI, SCAMPI
 - Agile Processes (Scrum, XP, ...), Unified Process, Object-Oriented Modeling, ...



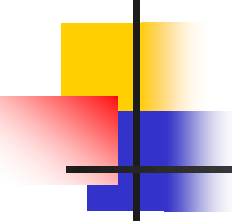
Introduction

- The SEI Software Capability Maturity Model (CMM) was built to provide guidance to the military services in selecting capable software contractors.
- The resulting method for evaluating their strengths and weaknesses has proved valuable for assessing other software organizations.



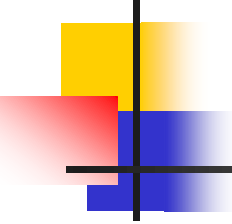
Introduction

- The textbook consists of 5 parts, each of which corresponds to a stage of SW process improvement.
 - Part 1: Software Process Maturity
 - Part 2: The Repeatable Process
 - Part 3: The Defined Process
 - Part 4: The Managed Process
 - Part 5: The Optimizing Process



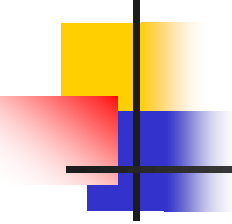
Introduction

- Part 1: Software Process Maturity
 - Describes a framework for software process management, the use of this framework in process assessment, and the steps required to initiate effective software process change
- Part 2: The Repeatable Process
 - Outlines the actions needed to establish basic control over the software process.
 - These provide the stability needed for orderly and continuous process improvement.



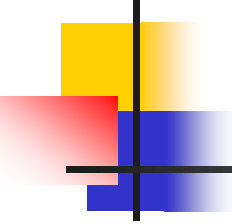
Introduction

- Part 3: The Defined Process
 - Describes **how to specify the development process and the technical and managerial concepts** needed to control it.
 - Software development is a surprise-prone business, and a defined process permits **more orderly reaction** to unanticipated events.



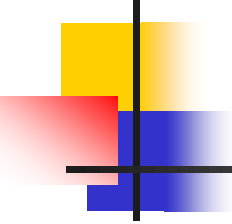
Introduction

- Part 4: The Managed Process
 - The method of **quantitative software process control** are introduced.
 - At this stage data is gathered and analyzed to support **quantitative quality and process management**.
 - These are the tools that can guide us to a better understanding of our work, a more precise way to control our actions, and a truly informed basis for **sustained process improvement**.



Introduction

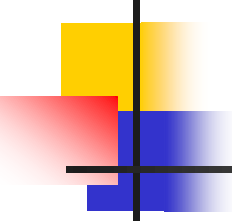
- Part 5: The Optimizing Process
 - Presents the final stage of software process evolution; it shifts the focus from fixing problems to **preventing** them.
 - At this level software managers and professionals learn to **use quantitative process methods** to harness technology for continuing quality and productivity improvement.



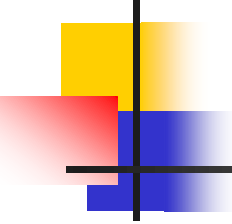
A Software Maturity Framework

- The software process is the set of tools, methods, and practices we use to produce a software product.
 - What
 - Who
 - When

A Software Maturity Framework

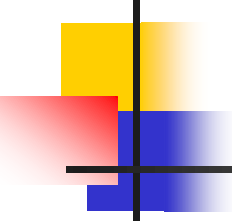


- **Practice** is a broad array of **concepts, principles, methods, and tools** that you must consider as SW is planned and developed [Pressman 2005].
 - It represents the **details – the technical considerations and how to's** – that are below the surface of the SW process: the things that you'll need to actually build high-quality SW.
 - What is the **work product**?
 - **Practice** encompasses the technical activities that produce all **work products** that are defined by the SW process model that has been chosen.



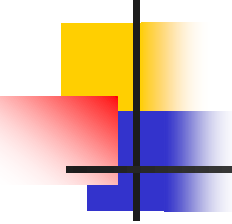
A Software Maturity Framework

- The characteristics of a truly effective software process
 - Fundamentally, it must be predictable.
 - That is, cost estimates and schedule commitments must be met with reasonable consistency, and the resulting products should generally meet users' functional and quality expectations.



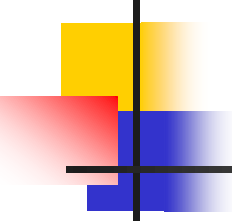
A Software Maturity Framework

- The objectives of software process management are to produce products according to plan while simultaneously improving the organization's capability to produce better products.



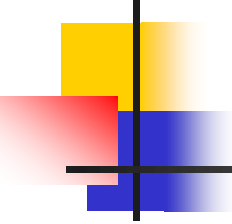
A Software Maturity Framework

- The basic principles are those of **statistical process control**, which have been used successfully in many fields.
- A process is said to be **stable or under statistical control** if its future performance is **predictable within established statistical limits**.
- When a process is under **statistical control**, repeating the work in roughly the same way will produce roughly the same result.



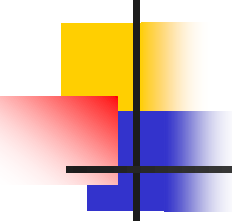
A Software Maturity Framework

- To obtain consistently better results, it is thus necessary to improve the process.
- If the process is not under statistical control, sustained progress is not possible until it is.
- The basic principle behind statistical control is measurement.



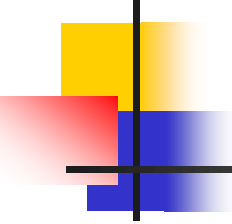
A Software Maturity Framework

- There are several factors to consider in **measuring** the process.
 - The numbers must **properly represent** the process being controlled.
 - The numbers must be sufficiently well defined and verified to provide a reliable basis for action.
 - The mere act of measuring human processes **changes** them.
 - It is thus essential to limit the measurements to those with a predefined use.



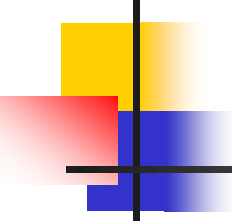
Software Process Improvement

- The improvement of software development organizations follows six steps:
 - Understand the current status of the development process.
 - Develop a vision of the desired process.
 - Establish a list of required process improvement actions in order of priority.
 - Produce a plan to accomplish the required actions.
 - Commit the resource to execute the plan.
 - Start over at step 1.



Process Maturity Levels

- Five levels of process maturity
 - Initial
 - Repeatable
 - Basic Management Control
 - Defined
 - Process Definition
 - Managed
 - Process Measurement
 - Optimizing
 - Process Control



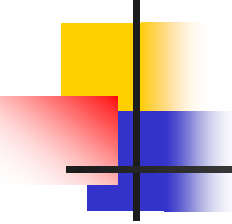
Process Maturity Levels

- CMM

- Initial
- Repeatable
- Defined
- Managed
- Optimizing

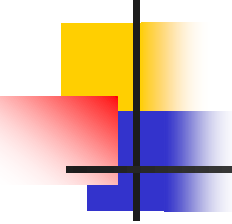
- CMMI

- Initial
- Managed
- Defined
- Quantitatively Managed
- Optimizing



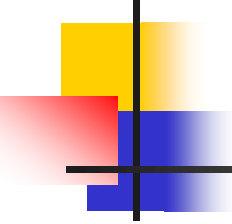
Process Maturity Levels

- Initial (Level 1)
 - Until the process is under statistical control, orderly progress in process improvement is not possible.
 - While there are many degrees of statistical control, the first step is to achieve rudimentary predictability of schedules and costs.
- Repeatable (Level 2)
 - The organization has achieved a stable process with a repeatable level of statistical control by initiating rigorous project management of commitments, cost, schedules, and changes.



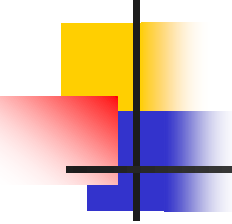
Process Maturity Levels

- Defined (Level 3)
 - The organization has **defined the process** as a basis for consistent implementation and better understanding.
 - At this point advanced technology can usefully be introduced.
- Managed (Level 4)
 - The organization has initiated **comprehensive process measurements and analysis**.
 - This is when **the most significant quality improvements begin**.



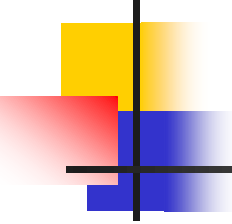
Process Maturity Levels

- Optimizing (Level 5)
 - The organization now has a foundation for continuing improvement and optimization of the process.



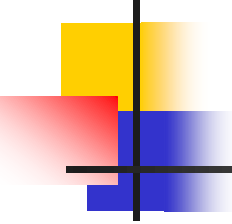
Process Maturity Levels

- Process maturity levels have been selected because they:
 - Reasonably represent the actual historical phases of evolutionary improvement of real SW organizations
 - Represent a measure of improvement that is reasonable to achieve from the prior level
 - Suggest interim improvement goals and progress measures
 - Make obvious a set of immediate improvement priorities, once an organization's status in this framework is known



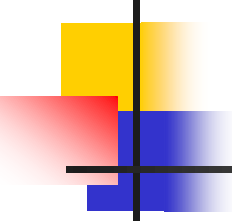
The Initial Process (Level 1)

- Ad hoc
- Often even chaotic
- At this stage the organization typically operate without formalized procedures, cost estimates, and project plans.
- Tools are neither well integrated with the process nor uniformly applied.
- Change control is lax, and there is little senior management exposure or understanding of the problems and issues.



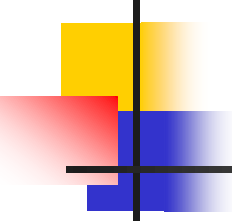
The Initial Process (Level 1)

- While organizations at this level may have formal procedures for planning and tracking their work, there is **no management mechanism** to ensure that they are used.
- The best test is to observe how such an organization behaves in a **crisis**.
 - If it abandons established procedures and essentially reverts to coding and testing, it is likely to be at the Initial Process Level.



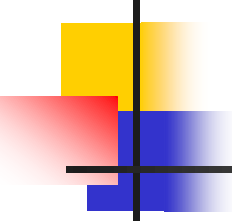
The Initial Process (Level 1)

- Organizations at the Initial Process Level can improve their performance by instituting basic project controls.
 - Project management
 - Management oversight
 - Quality assurance
 - Change control



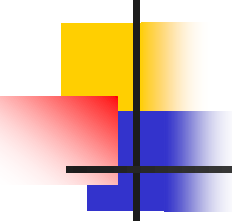
The Initial Process (Level 1)

- Project management
 - The fundamental role of a project management system is to ensure effective control of commitments.
 - Adequate preparation, clear responsibility, a public declaration, a dedication to performance
 - Starts with an understanding of the job's magnitude
 - In any projects, a plan must be developed to determine the best schedule and the anticipated resources required.



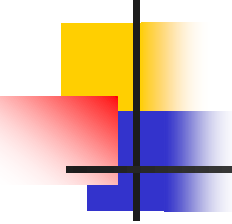
The Initial Process (Level 1)

- Management oversight
 - A suitably disciplined SW development organization must have senior management oversight.
 - Review and approval of all major development plans prior to their official commitment
 - A quarterly review should be conducted of facility-wide process compliance, installed quality performance, schedule tracking, cost trends, computing service, and quality and productivity goals by project.
 - The lack of such reviews typically results in uneven and generally inadequate implementation of the process as well as frequent over-commitments and cost surprise.



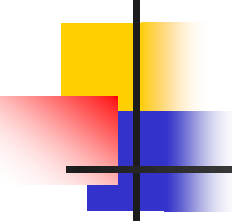
The Initial Process (Level 1)

- Quality assurance
 - A quality assurance group is charged with assuring management that SW work is done the way it is supposed to be done.
 - To be effective, the assurance organization must have an independent reporting line to senior management and sufficient resources to monitor performance of all key planning, implementation, and verification activities.
 - This generally requires an organization of about 3 percent to 6 percent the size of the SW organization.



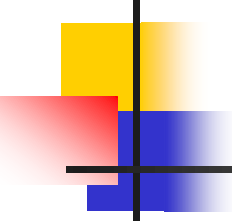
The Initial Process (Level 1)

- Change control
 - Change control for SW is fundamental to business and financial control as well as to technical stability.
 - To develop quality SW on a predictable schedule, **requirements must be established and maintained with reasonable stability** throughout the development cycle.
 - Requirement changes; design and code changes
 - If changes are not controlled, then orderly design, implementation, and test is impossible and no quality plan can be effective.



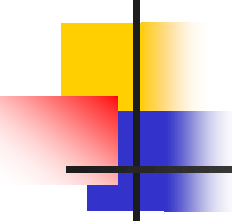
The Repeatable Process (Level 2)

- The Repeatable Process provides control over the way the organization establishes its plans and commitment.
- Organizations at this level face major risks when they are presented with new challenges.
 - New tools and methods will affect the process.
 - When the organization must develop a new kind of product, it is entering new territory.
 - Major organizational changes can also be highly disruptive.



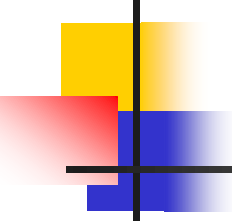
The Repeatable Process (Level 2)

- Key actions required to advance from the Repeatable to the next stage, the Defined Process, are to
 - Establish a process group (EPG, SEPG)
 - Establish a software development process architecture (or development life cycle)
 - Introduce a family of SE methods and technologies



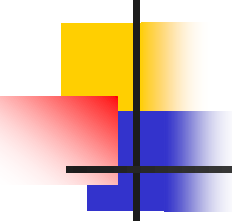
The Repeatable Process (Level 2)

- Key actions required to advance to the next stage:
 - Establish a process group
 - A process group is a technical resource that focuses exclusively on improving the SW process. (EPG, SEPG)
 - In SW organizations at early maturity levels, all the people are generally devoted to product work.
 - Until some people are given full-time assignments to work on the process, little orderly progress can be made in improving it.



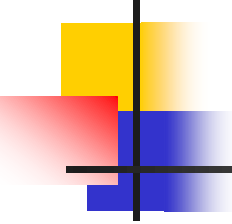
The Repeatable Process (Level 2)

- Key actions required to advance to the next stage:
 - Establish a SW development process architecture, or development life cycle, that describes the technical and management activities required for proper execution of the development process
 - The process architecture is a structural decomposition of the development life cycle into tasks, each of which has a defined set of prerequisites, functional descriptions, verification procedures, and task completion specifications.
 - The decomposition continues until each defined task is performed by an individual or single management unit.



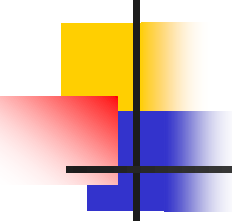
The Repeatable Process (Level 2)

- Key actions required to advance to the next stage:
 - If they are not already in place, introduce a family of SW engineering methods and technologies.
 - These include design and code inspections, formal design methods, library control systems, and comprehensive testing methods.
 - Prototyping should also be considered, together with the adoption of modern implementation languages.



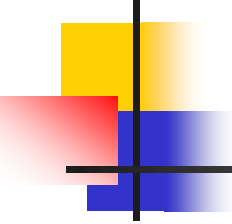
The Defined Process (Level 3)

- With the Defined Process, the organization has achieved the foundations for major and continuing progress.
- For example, the software team, when faced with a crisis, will likely continue to use the process that has been defined.
- The foundation has now been established for examining the process and deciding how to improve it.



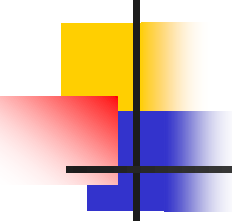
The Defined Process (Level 3)

- The key steps required to advance from the Defined Process to the next level:
 - Establish a minimum basic set of process measurements to identify the quality and cost parameters of each process step.
 - Establish a process database and the resources to manage and maintain it.
 - Provide sufficient process resources to gather and maintain this process data and to advise project members on its use.
 - Assess the relative quality of each product and inform management where quality targets are not being met.



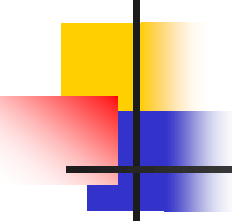
The Managed Process (Level 4)

- In advancing from the previous levels to the Managed Process, organizations should expect to make **substantial quality improvement**.
- The greatest problem with the Managed Process is **the cost of gathering data**.
- There are **an enormous number of potentially valuable measures** of the software process, but such data is expensive to gather and to maintain.



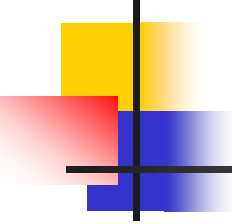
The Managed Process (Level 4)

- The two fundamental requirements for advancing from the Managed Process to the next level are:
 - Support automatic gathering of process data.
 - Use process data both to analyze and to modify the process to prevent problems and improve efficiency.



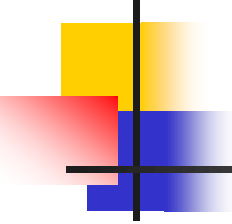
The Optimizing Process (Level 5)

- In varying degree, **process optimization** goes on at all levels of process maturity.
- With the step from the Managed Process to the Optimizing Process, however, there is **a paradigm shift**.
- **Up to this point** SW development managers have largely **focused on their products** and will typically gather and analyze only data that directly relates to **product improvement**.



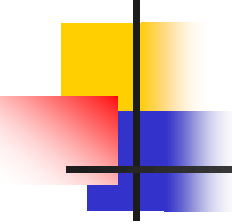
The Optimizing Process (Level 5)

- In the Optimizing Process, the data is available to tune the process itself.
- With a little experience, management will soon see that process optimization can produce major quality and productivity benefits.
- With the Optimizing Process, the organization has the means to identify the weakest elements of the process and to fix them.



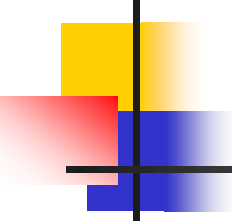
People in the Optimizing Process

- Clearly any SW process is **dependent on the quality of the people** who implement it.
- The Optimizing Process enhances **the talent of quality people** in several ways.
 - It helps managers understand **where help is needed and how best to provide people with the support they require.**
 - It lets the professionals communicate in concise, quantitative terms.
 - It provides a framework for the professionals to understand their work performance and to see how to improve it.



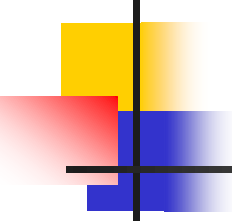
The Need for the Optimizing Process

- Unless we dramatically improve our error rates, **the greater volume of code** will mean **increased risk of error**.
- **The complexity of our systems** is increasing, which will make the systems progressively more **difficult to test**.
- As well as being a management issue, **quality** is an economic one.
 - It is always possible to do more tests, but it **costs time and money** to do so.



The Need for the Optimizing Process

- It is only with the Optimizing Process that the data is available to understand the costs and benefits of such work.
- The Optimizing Process provides the foundation for significant advances in SW quality and simultaneous improvements in productivity.

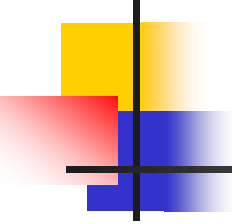


The Need for the Optimizing Process

- Transition from Level 1 to Level 2 or from Level 2 to Level 3 take from one to three years, even with a dedicated management commitment to process improvement [Humphrey 1989].

Process Levels Comparison

	CMM Maturity	CMMI Maturity	CMMI v1.2 Capability	CMMI v1.3 Capability	ISO 15504 Capability
Level 5	Optimizing	Optimizing	Optimizing	-	Optimizing
Level 4	Managed	Quantitatively Managed	Quantitatively Managed	-	Predictable
Level 3	Defined	Defined	Defined	Defined	Established
Level 2	Repeatable	Managed	Managed	Managed	Managed
Level 1	Initial	Initial	Performed	Performed	Performed
Level 0	-	-	Incomplete	Incomplete	Incomplete



References

- Humphrey, Watts S., *Managing the Software Process*, The SEI Series in Software Engineering, Addison-Wesley, 1989. (29th Printing, May 2003) (ISBN 0-201-18095-2)